

Disproof of TLMC Conjecture 00000000340

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Abstract

Verdict: **FALSE**. The conjecture asserts that for every algebraic irrational α the geometric means of continued-fraction partial quotients, $G_N(\alpha) = (\prod_{j \leq N} a_j)^{1/N}$, satisfy $\liminf_{N \rightarrow \infty} G_N(\alpha) = 1$ and $\limsup_{N \rightarrow \infty} G_N(\alpha) = \infty$. The counterexample $\alpha = \sqrt{2} = [1; 2, 2, 2, \dots]$ is algebraic irrational, yet all its partial quotients after the first equal 2, so $G_N(\sqrt{2}) = 2^{(N-1)/N} \rightarrow 2$. The sequence is confined to $[\sqrt{2}, 2)$, whence $\liminf G_N = \limsup G_N = 2$: both assertions fail simultaneously.

1 The conjecture

00000000340. Let $G_N(\alpha) = (\prod_{j \leq N} a_j)^{1/N}$, where $a_1, a_2, \dots$ are the partial quotients of the simple continued fraction of α . Conjecture: for every algebraic irrational α ,

$$\liminf_{N \rightarrow \infty} G_N(\alpha) = 1 \quad \text{and} \quad \limsup_{N \rightarrow \infty} G_N(\alpha) = \infty.$$

2 Counterexample: $\alpha = \sqrt{2}$

The regular continued fraction of $\sqrt{2}$ is

$$\sqrt{2} = [1; 2, 2, 2, \dots],$$

i.e. $a_1 = 1$ and $a_j = 2$ for every $j \geq 2$. This is exact, not asymptotic: writing the standard integer CF recurrence for $\sqrt{D}$ ($m \leftarrow da - m$, $d \leftarrow (D - m^2)/d$, $a \leftarrow \lfloor (\lfloor \sqrt{D} \rfloor + m)/d \rfloor$) with $D = 2$ gives $(m, d) = (0, 1)$, then the period $(1, 2)$, and one checks $p_k^2 - 2q_k^2 = (-1)^{k+1}$ for the convergents (Pell's equation), which forces $a_j = 2$ for all $j \geq 2$.

Therefore, for every $N \geq 1$,

$$\prod_{j \leq N} a_j = 1 \cdot 2^{N-1} = 2^{N-1}, \quad G_N(\sqrt{2}) = (2^{N-1})^{1/N} = 2^{(N-1)/N} = 2 \cdot 2^{-1/N}.$$

3 Why both claims fail

The sequence $G_N(\sqrt{2}) = 2^{(N-1)/N}$ is strictly increasing (for $N \geq 1$) and converges to 2. Consequently

$$\liminf_{N \rightarrow \infty} G_N(\sqrt{2}) = \limsup_{N \rightarrow \infty} G_N(\sqrt{2}) = 2,$$

so

N	$G_N(\sqrt{2}) = 2^{(N-1)/N}$
10	1.866065983
100	1.986184991
1000	1.998614186

Table 1: Exact values $G_N(\sqrt{2}) = 2^{(N-1)/N}$; the limit is 2.

- $\limsup G_N = \infty$ **fails**: $G_N \leq 2$ for every N (indeed $G_N < 2$);
- $\liminf G_N = 1$ **fails**: $G_N \geq \sqrt{2} > 1$ for every $N \geq 2$; more sharply, $G_N \geq 1.99$ for every $N \geq 139$.

All inequalities used above are provable in elementary integer arithmetic; e.g. $G_N \geq 1.99$ is the integer inequality $199^N \leq 2^{N-1} \cdot 100^N$, and the threshold $N = 139$ is *sharp*: at $N = 138$ the inequality fails ($199^{138} > 2^{137} \cdot 100^{138}$). At $N = 1000$: $199^{1000} \leq 2^{999} \cdot 100^{1000} \leq 201^{1000}$, i.e. $1.99 \leq G_{1000} \leq 2.01$, and in fact $G_{1000} = 1.998614186\dots$

These statements are certified in machine-checked form in `lean4/Main.lean` (core Lean 4 only, no Mathlib, and *zero axioms*: every proof is by `rfl/decide` and explicit N-lemmas; `Check.lean` audits each theorem with `#print axioms`).

4 Boundary of the disproof

- **The failure is structural, not an $\sqrt{2}$ accident.** Every quadratic irrational has an eventually periodic continued fraction, hence bounded partial quotients and bounded (G_N) ; moreover G_N is bounded below by a constant > 1 (the geometric mean of one period). So the conjecture fails on the whole class of quadratic irrationals (indeed on all badly approximable numbers).
- **The almost-everywhere version fails too.** For Lebesgue-a.e. α , Birkhoff’s ergodic theorem applied to the Gauss measure $d\mu = \frac{1}{\ln 2} \frac{dx}{1+x}$ gives $\frac{1}{N} \sum_{j \leq N} \ln a_j \rightarrow \int \ln a \, d\mu = \ln K$, so $G_N(\alpha) \rightarrow K \approx 2.685452001$ (Khinchin’s constant). Thus $\limsup G_N = \infty$ fails for a.e. α as well; the conjecture cannot be rescued by replacing “algebraic” with “almost every”.
- **What is not settled.** Whether some non-quadratic algebraic irrational has unbounded partial quotients (e.g. for $\alpha = \sqrt[3]{2}$) is a classical open problem. The present disproof does not touch it; it only shows that the conjecture *as stated* is false, because a single algebraic irrational ($\sqrt{2}$, or any quadratic irrational) refutes it.

5 Reproduction

- `python3 reproduce.py` — recomputes the partial quotients of $\sqrt{2}$ with the exact integer CF recurrence, verifies $\prod_{j \leq N} a_j = 2^{N-1}$ for $N \leq 2000$, the values in the table, the bounds $G_N \leq 2$ and $G_N^2 \geq 2$, and the sharp threshold 139; all checks pass.
- `cd lean4 && lake build && lake env lean Check.lean` — builds the Lean certificate (toolchain `leanprover/lean4:v4.33.1`, no dependencies) and prints the axiom audit: every theorem “does not depend on any axioms”.

- `tectonic main.tex --outdir build` — rebuilds this PDF.